

Generative Computer Vision Models Project

- Prof. Dr.-Ing. Margret Keuper



PROJECT AGENDA

AIM : To utilize the principles and techniques discussed during the lecture on generation models to generate novel data samples.

WHAT TO DO: Build a new model (Algorithm)/ use a variant of existing generative models to tackle the task of generation.

BENEFITS: Self-Learning, Theory meets practical, Presentation skills.

PROJECT INSPIRATION

How to get inspired:

Computer Vision or ML Conferences

- **CVPR**: IEEE Conference on Computer Vision and Pattern Recognition.
- **ICCV**: International Conference on Computer Vision.
- **ECCV**: European Conference on Computer Vision.
- **NeurIPS**: Conference on Neural Information Processing Systems.
- **ICLR**: International Conference on Learning Representations.
- **ICML**: International Conference on Machine Learning.

PROJECT INSPIRATION

How to get inspired:

Computer Vision or ML Journals

- **Nature:** Multidisciplinary science journal featuring top AI work.
- **IJCV:** International Journal of Computer Vision.
- **TMLR:** Transactions on Machine Learning Research.
- **JMLR:** Journal of Machine Learning Research.
- **TPAMI:** IEEE Transactions on Pattern Analysis and Machine Intelligence.

CV DATASETS

- Meta Pointer: A large collection organized by CV Datasets. Yet another Meta pointer.
- ImageNet: a large-scale image dataset for visual recognition organized by WordNet hierarchy.
- SUN Database: a benchmark for scene recognition and object detection with annotated scene categories and segmented objects.
- Places Database: a scene-centric database with 205 scene categories and 2.5 millions of labelled images

CV DATASETS

- Microsoft COCO: a new benchmark for image recognition, segmentation and captioning.
- Flickr100M: 100 million creative commons Flickr images.
- Labeled Faces in the Wild: a dataset of 13,000 labeled face photographs.
- Human Pose Dataset: a benchmark for articulated human pose estimation.
- YouTube Faces DB: a face video dataset for unconstrained face recognition in videos.

PROJECT TOPICS

Generative Models discussed in lecture (Below) :

- Normalizing Flow Models
- Diffusion Models
- Flow Matching Models

Choose any of the above generative Models for the Project.

PROJECT WORKFLOW

- Work Individually/ Groups of two.
- Choose a Generative method you will be investigating.
- Build a generative model for the task of generation (CIFAR10 Dataset).
- What paper(s) do you need to read?.
- How do you evaluate the results? (Ex : FID score).
- Prepare a 5 minute presentation to present your findings.
- Presentations are on May 18, 2026

PRESENTATION INSTRUCTIONS

- **Introduction (15 sec)**

Motivation or goal of the work / papers read or selected.

- **Chosen Generative Model & How It Works (2.25 min ± 15)**

Name the model (e.g., Diffusion, Normalizing Flow or Flow Matching).

High-level diagram or animation.

Mention core idea.

- **Methodology (1 min ± 15)**

Dataset used.

Architecture tweaks (if any).

Training Summary.

PRESENTATION INSTRUCTIONS

- **Visual Results + Quantitative Analysis (1min)**

Side-by-side comparison of generated vs real samples.

FID score or any other metric.

Keep plots if any concise, focus on readability.

- **Conclusion & Limitations (40 sec)**

Brief summary and Limitations or failure cases (briefly).

- **Please ensure your presentation is approximately 5 minutes, with a flexibility of ± 15 seconds.**

- **Note: Keep slides concise — avoid overcrowding them with text.**

Github Repositories

- **Normalizing Flows**

RealNVP : <https://github.com/xqding/RealNVP>

Glow : https://github.com/Naagar/Glow_NormalizingFlow_Implementation

Residual Flows : <https://github.com/rtqichen/residual-flows>

Normalizing Flows : <https://github.com/VincentStimper/normalizing-flows>

- **Diffusion**

DDPM : <https://github.com/zoubohao/DenoisingDiffusionProbabilityModel-ddpm->

Conditional Diffusion Model : https://github.com/fastai/fastdiffusion/blob/master/nbs/tcapelle/Diffusion_models_with_fastai_conditional_cifart_EMA.ipynb

- **Flow Matching**

https://virtualhumanlab.io/EN_blog_FromNoisetolImages

Thank you :) !

Please contact tejaswini.medi@uni-mannheim.de for any further queries .